

## 34. Hypercomplex Quantities and Representation Theory

*Continuation: Chapter I, §§5–7, and Chapter II, §§8–14.*

### § 5. Completely reducible groups

A group is called *completely reducible* if it is the direct product of finitely many simple groups:

$$\mathfrak{G} = \mathfrak{A}_1 \times \cdots \times \mathfrak{A}_n.$$

In this case the groups

$$\mathfrak{A}_1 \times \cdots \times \mathfrak{A}_n \supset \mathfrak{A}_1 \times \cdots \times \mathfrak{A}_{n-1} \supset \cdots \supset \mathfrak{A}_1 \supset \mathfrak{E}$$

form a composition series. Its length is  $n$ , and its composition factors are

$$\sim \mathfrak{A}_n, \mathfrak{A}_{n-1}, \dots, \mathfrak{A}_1.$$

*If  $\mathfrak{G}$  is completely reducible, then every normal divisor is a direct factor, and the other factor can be chosen as a product of such simple groups as occur in a prescribed product decomposition of  $\mathfrak{G}$ . The normal divisor  $\mathfrak{H}$  is itself completely reducible.*

*Proof.* We have

$$\mathfrak{G} = \mathfrak{H} \mathfrak{A}_1 \mathfrak{A}_2 \cdots \mathfrak{A}_n. \quad (3)$$

Put  $\mathfrak{H}_i = \mathfrak{H} \mathfrak{A}_1 \cdots \mathfrak{A}_i$ . Then  $\mathfrak{H}_{i+1} = \mathfrak{H}_i \mathfrak{A}_{i+1}$ . Either  $\mathfrak{A}_{i+1} \leq \mathfrak{H}_i$ , whence  $\mathfrak{H}_{i+1} = \mathfrak{H}_i$ ; in that case we omit the factor  $\mathfrak{A}_{i+1}$  from (3). Or  $\mathfrak{H}_i \cap \mathfrak{A}_{i+1}$  is a proper normal divisor of  $\mathfrak{A}_{i+1}$ , hence is  $\mathfrak{E}$ , and then  $\mathfrak{H}_i \times \mathfrak{A}_{i+1}$  is direct. After the superfluous factors have been omitted, (3) becomes

$$\mathfrak{G} = \mathfrak{H} \times \mathfrak{A}_{i_1} \times \cdots \times \mathfrak{A}_{i_r},$$

so that  $\mathfrak{H}$  is a direct factor. Further,

$$\mathfrak{H} \sim \mathfrak{G} / (\mathfrak{A}_{i_1} \times \cdots \times \mathfrak{A}_{i_r}) \sim \mathfrak{A}_{j_1} \times \cdots \times \mathfrak{A}_{j_\mu},$$

where  $\mathfrak{A}_{j_1}, \dots, \mathfrak{A}_{j_\mu}$  are the remaining  $\mathfrak{A}_i$ . Thus  $\mathfrak{H}$  is completely reducible.

*Corollary. Every decomposition of a completely reducible group into directly indecomposable factors is a decomposition into simple factors, and therefore gives rise to a composition series; from this follows the unique determination of the factors up to isomorphism.*

### § 6. Modules with respect to a field. Hypercomplex systems

An almost trivial example of completely reducible groups with operators is given by modules with finite basis with respect to a not necessarily commutative field.

Let  $\mathfrak{G}$  be a right  $K$ -module, and let the identity element  $e$  of  $K$  be at the same time the identity operator:  $ae = a$  for  $a$  in  $\mathfrak{G}$ . Every submodule  $aK$  derived from an  $a$  is simple, namely equal to the module derived from any one of its non-zero elements. Thus if  $\mathfrak{H}$  is the submodule derived from some elements, and  $a$  is an element not belonging to  $\mathfrak{H}$ , then  $\mathfrak{H} \cap aK = \mathfrak{E}$ , so  $\mathfrak{H} + aK$  is direct. Starting with any basis element  $a_1$ , one can therefore keep adjoining new basis elements  $a_2, a_3, \dots$ , and obtains  $\mathfrak{G}$  as the direct sum

$$\mathfrak{G} = a_1 K + a_2 K + \cdots + a_n K.$$

*Thus  $\mathfrak{G}$  is completely reducible.*

The number  $n$ , the length of the composition series, is called the *rank* of  $\mathfrak{G}$  with respect to  $K$ . Because of the uniqueness of the representation of the elements of  $\mathfrak{G}$  (and because  $e$  has been assumed to be the identity operator), the basis  $a_1, \dots, a_n$  is linearly independent.

Every submodule  $\mathfrak{H}$  is a direct summand:

$$\mathfrak{G} = \mathfrak{H} + \mathfrak{R} = (h_1 K + \cdots + h_s K) + (r_1 K + \cdots + r_{n-s} K),$$

or: *every linearly independent basis of  $\mathfrak{H}$  can be completed to a linearly independent basis of  $\mathfrak{G}$ . The  $r_i$  may even be chosen from the original basis elements  $a_i$ .*

Let  $(a_1, \dots, a_n)$  and  $(c_1, \dots, c_n)$  be linearly independent bases. Then

$$c_k = \sum_i a_i \pi_{ik},$$

or, written in matrices,

$$(c_1, \dots, c_n) = (a_1, \dots, a_n)P, \quad P = \begin{pmatrix} \pi_{11} & \cdots & \pi_{1n} \\ \vdots & & \vdots \\ \pi_{n1} & \cdots & \pi_{nn} \end{pmatrix}.$$

Conversely,

$$(a_1, \dots, a_n) = (c_1, \dots, c_n)Q.$$

It follows that

$$(a_1, \dots, a_n) = (a_1, \dots, a_n)PQ, \quad (c_1, \dots, c_n) = (c_1, \dots, c_n)QP,$$

and, since the  $a_i$  and  $c_i$  are linearly independent,

$$PQ = QP = E.$$

Special  $K$ -modules that are also rings are the “hypercomplex systems.”

A ring  $\mathfrak{o}$  which is at the same time a right module with respect to a commutative field  $K$  is called a *hypercomplex system with respect to  $K$*  (also called an “algebra over  $K$ ” in the literature) if:

1. its rank is finite (let a linearly independent basis be  $u_1, \dots, u_n$ );
2.  $ab \cdot x = a \cdot bx = ax \cdot b$ . This is expressed by saying that  $K$  is joined commutatively with  $\mathfrak{o}$ ;
3. the identity element  $e$  of  $K$  is also the identity operator:  $ae = a$  for  $a$  in  $\mathfrak{o}$ .

Since

$$\left(\sum_i u_i \alpha_i\right) \left(\sum_k u_k \beta_k\right) = \sum_{i,k} (u_i u_k) (\alpha_i \beta_k),$$

the system is uniquely determined once, in addition to  $K$ , the *multiplication table* is given, i.e. once it is known how each product  $u_i u_k$  is expressed through the  $u_j$ :

$$u_i u_k = \sum_j u_j \gamma_{ik}^j.$$

The  $\gamma_{ik}^j$  must satisfy the familiar relations following from the associative law.

If  $\mathfrak{o}$ , as we shall often assume, has an identity element  $e$  ( $ea = ae = a$  for all  $a$ ), then the elements  $x$  of  $K$  may be identified with  $ex$ , and  $K$  may be regarded as a subfield of  $\mathfrak{o}$ . The hypercomplex system may then also be described as a *ring of finite rank with respect to a field lying in the center*.

An example of a hypercomplex system is the *group ring* of a finite group: one takes the group elements as the basis elements  $u_i$ , and group multiplication as multiplication. The field  $K$  is arbitrary.

## § 7. Matrices

The square matrices  $(\pi_{ik})$ , with  $\pi_{ik}$  from a ring  $\mathfrak{o}$ , form a ring under the ordinary matrix multiplication  $\sum_j \pi_{ij} \rho_{jk} = \sigma_{ik}$  and addition  $\pi_{ik} + \rho_{ik} = \sigma_{ik}$ .

A matrix with elements from a field  $K$  for which there is both a right inverse and a left inverse is called *regular*.

We saw in §6: *The transition matrix between two linearly independent bases of a  $K$ -module is regular.*

*For regularity a right inverse suffices.* Indeed, form a right  $K$ -module with linearly independent basis  $(a_1, \dots, a_n)$  (the module of linear forms in the indeterminates  $a_1, \dots, a_n$ ), and put

$$(c_1, \dots, c_n) = (a_1, \dots, a_n)P.$$

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<sup>0</sup>Condition 2 says that multiplication by an element of  $K$  is a homomorphism for  $\mathfrak{o}$  when  $\mathfrak{o}$  is regarded both as an  $\mathfrak{o}$ -left-module and as an  $\mathfrak{o}$ -right-module. By virtue of 3. these homomorphisms are even isomorphisms.

Then

$$(c_1, \dots, c_n)P^{-1} = (a_1, \dots, a_n)PP^{-1} = (a_1, \dots, a_n),$$

so  $(c_1, \dots, c_n)$  is again a basis, hence again linearly independent, and the matrix  $P$  is regular. Moreover the right inverse is equal to the left inverse.

Similarly, a left inverse suffices.

*If  $P$  has a left inverse, then  $P$  is not a left zero divisor. Proof.* From  $PA = 0$  follows  $A = P^{-1}PA = 0$ .

Thus  $P$  also has only one right inverse, which by §6 coincides with the left inverse.

If, as usual,  $P_x$  denotes the matrix  $(\pi_{ik}x)$ , and  $C_{ik}$  denotes the matrix which has the identity element in the  $ik$  place and zeros everywhere else, then

$$(\pi_{ik}) = \sum_{i,k} C_{ik}\pi_{ik};$$

so the matrices over  $K$  form a  $K$ -module of rank  $n^2$ . The  $C_{ik}$  satisfy the relations

$$C_{ij}C_{jk} = C_{ik}, \quad C_{ij}C_{\bar{j}k} = 0, \quad j \neq \bar{j}.$$

Thus the matrices over  $K$  form a finite  $K$ -module and, if  $K$  is commutative, a hypercomplex system.

## Chapter II. Non-commutative ideal theory

### § 8. Homomorphism theorem for rings

If  $\mathfrak{a}$  is a two-sided ideal in  $\mathfrak{o}$ , then the residue-class domain  $\mathfrak{o}/\mathfrak{a}$  is not only an  $\mathfrak{o}$ -module but also a ring, as is easily seen. Every homomorphic image of  $\mathfrak{o}$  is again a ring, and is isomorphic to a residue-class ring  $\mathfrak{o}/\mathfrak{a}$ , where  $\mathfrak{a}$  is a two-sided ideal. The proof is as for groups.

In particular, if  $\mathfrak{o}$  is a field, then there are no ideals other than  $(0)$  and  $\mathfrak{o}$ ; hence here the homomorphism is either an isomorphism, or assigns zero to every element.

In particular, suppose the ring  $\mathfrak{o}$  is a right  $K$ -module, where  $K$  is to be a ring elementwise commuting with  $\mathfrak{o}$ :

$$ab \cdot x = a \cdot bx = ax \cdot b$$

(for example, in the case of a hypercomplex system with respect to a commutative field  $K$ ). Then as admissible right, left, or two-sided ideals one considers only those which are at the same time  $K$ -modules, and in addition to ring homomorphism one also requires operator homomorphism with respect to multiplication by  $K$ . The admissible ideals become bimodules with respect to  $\mathfrak{o}$  and  $K$ . (For hypercomplex systems, by “ideals” without further qualification one always means only those admissible with respect to the underlying coefficient field  $K$ .) In this enlarged sense too, the homomorphism theorem above remains valid.

An example of ring homomorphism is the transition from a multiplier domain  $\mathfrak{o}$  to the absolute multiplier domain (§1, Example 4). Thus the absolute multiplier domain is isomorphic to the residue-class ring  $\mathfrak{o}/\mathfrak{d}$ , where  $\mathfrak{d}$  is the two-sided ideal of elements of  $\mathfrak{o}$  which annihilate  $\mathfrak{M}$ .

From the validity of the homomorphism theorem there follow, as in §2, the first and second isomorphism theorems for ring and operator isomorphism.

All products  $\mathfrak{A}\mathfrak{B}$  of subsets of  $\mathfrak{o}$  are to be understood as module products:  $\mathfrak{A}\mathfrak{B}$  is the totality of all sums  $\sum ab$ , with  $a$  in  $\mathfrak{A}$ ,  $b$  in  $\mathfrak{B}$ , and  $a\mathfrak{B}$  is the totality of all sums  $\sum ab$ ,  $b$  in  $\mathfrak{B}$ . If  $\mathfrak{B}$  is a right module with respect to some ring as operator domain, then the product  $\mathfrak{A}\mathfrak{B}$ , respectively  $a\mathfrak{B}$ , is also a right module. In particular: a right ideal, an admissible ideal with respect to a coefficient domain  $K$ , etc. If  $(\mathfrak{A}, \mathfrak{B})$  denotes the sum of the modules  $\mathfrak{A}$  and  $\mathfrak{B}$ , then the rules of calculation are

$$\mathfrak{A}\mathfrak{B} \cdot \mathfrak{C} = \mathfrak{A} \cdot \mathfrak{B}\mathfrak{C},$$

$$\mathfrak{A}(\mathfrak{B}, \mathfrak{C}) = (\mathfrak{A}\mathfrak{B}, \mathfrak{A}\mathfrak{C}), \quad (\mathfrak{A}, \mathfrak{B})\mathfrak{C} = (\mathfrak{A}\mathfrak{C}, \mathfrak{B}\mathfrak{C}).$$

The direct sum is denoted by  $\mathfrak{A} + \mathfrak{B}$  (§4).

In what follows we shall often consider rings satisfying a maximal or minimal condition (§3) for right ideals. Hypercomplex systems in particular belong to these, since by the convention just made only  $K$ -modules are admitted as ideals, and their rank therefore remains bounded (§6).

## § 9. Idempotent elements. Direct sum decomposition into right ideals

Let  $\mathfrak{o}$  be a ring with identity element.

If  $\mathfrak{r}$  is a right ideal, then  $\mathfrak{r}\mathfrak{o} \subseteq \mathfrak{r}$ , and because of the identity element even  $\mathfrak{r}\mathfrak{o} = \mathfrak{r}$ . Correspondingly for left ideals:  $\mathfrak{o}\mathfrak{l} = \mathfrak{l}$ .

An element  $c$  is called idempotent if  $c^2 = c$ .

If  $\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n$  is a decomposition into right ideals, and if

$$e = e_1 + \cdots + e_n, \quad e_i \in \mathfrak{r}_i,$$

then

$$e_i^2 = e_i, \quad e_i e_k = 0 \quad (i \neq k) \quad (\text{"orthogonality relations"}),$$

$$\mathfrak{r}_i = e_i \mathfrak{o}.$$

*Proof.* Let  $r$  be an element of  $\mathfrak{r}_1$ . Then

$$r = er = e_1 r + \cdots + e_n r, \quad e_i r \in \mathfrak{r}_i,$$

but also

$$r = r + 0 + \cdots + 0,$$

so

$$e_1 r = r, \quad e_i r = 0 \quad (i \neq 1).$$

From the first formula it follows that  $\mathfrak{r}_1 \subseteq e_1 \mathfrak{o}$ ; on the other hand  $e_1 \mathfrak{o} \subseteq \mathfrak{r}_1$ , so  $e_1 \mathfrak{o} = \mathfrak{r}_1$ . Specializing  $r = e_1$  gives

$$e_1^2 = e_1, \quad e_i e_1 = 0 \quad (i \neq 1).$$

The same holds when the index 1 is replaced by  $k$ .

*Conversely*, if  $e = \sum e_i$ ,  $e_i e_k = 0$  ( $i \neq k$ ),  $e_i^2 = e_i$ , and one puts  $\mathfrak{r}_i = e_i \mathfrak{o}$ , then  $\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n$ .

*Proof.* Every element  $r$  of  $\mathfrak{o}$  is

$$r = er = e_1 r + \cdots + e_n r,$$

hence

$$\mathfrak{o} = (e_1 \mathfrak{o}, \dots, e_n \mathfrak{o}).$$

But the representation is unique; for from

$$0 = e_1 a_1 + \cdots + e_n a_n$$

it follows, upon multiplication by  $e_i$ , that

$$e_i^2 a_i = e_i a_i = 0.$$

From a right representation

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n = e_1 \mathfrak{o} + \cdots + e_n \mathfrak{o}$$

there accordingly follows a left representation

$$\mathfrak{o} = \mathfrak{l}_1 + \cdots + \mathfrak{l}_n = \mathfrak{o} e_1 + \cdots + \mathfrak{o} e_n.$$

If the  $\mathfrak{r}_i$  are directly indecomposable, then so are the  $\mathfrak{l}_i$ . For a decomposition, say of  $\mathfrak{l}_1$ , would mean

$$\mathfrak{l}_1 = \mathfrak{o} e_1 = \mathfrak{o} e'_1 + \mathfrak{o} e''_1, \quad \mathfrak{o} = \mathfrak{o} e'_1 + \mathfrak{o} e''_1 + \mathfrak{o} e_2 + \cdots + \mathfrak{o} e_n.$$

From this would result the right decomposition

$$\mathfrak{o} = e'_1 \mathfrak{o} + e''_1 \mathfrak{o} + e_2 \mathfrak{o} + \cdots + e_n \mathfrak{o} = \mathfrak{r}'_1 + \mathfrak{r}''_1 + \mathfrak{r}_2 + \cdots + \mathfrak{r}_n,$$

$$\mathfrak{r}_1 \cong \mathfrak{o} / (\mathfrak{r}_2 + \cdots + \mathfrak{r}_n) = \mathfrak{r}'_1 + \mathfrak{r}''_1,$$

so  $\mathfrak{r}_1$  would be decomposable.

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<sup>0</sup>There is another converse: if  $e_i e_k = 0$ ,  $e_i^2 = e_i$ ,  $c = \sum e_i$ , then  $c$  is a left identity for the ring  $e_1 \mathfrak{o} + \cdots + e_n \mathfrak{o}$ . The fact that the sum is direct is seen in exactly the same way as above.

## § 10. Decomposition into two-sided ideals

Let  $\mathfrak{o}$  again be a ring with identity element, and let

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_n \quad (1)$$

be a decomposition of  $\mathfrak{o}$  into two-sided ideals. Then the orthogonality relations also hold for the ideals:

$$\mathfrak{a}_i \mathfrak{a}_k = 0 \quad (i \neq k), \quad \mathfrak{a}_i^2 = \mathfrak{a}_i. \quad (2)$$

*Proof.* Let  $\mathfrak{b}_1 = \mathfrak{a}_2 + \cdots + \mathfrak{a}_n$ ,  $\mathfrak{o} = \mathfrak{a}_1 + \mathfrak{b}_1$ . Then  $\mathfrak{a}_1 \mathfrak{b}_1 \leq \mathfrak{a}_1 \cap \mathfrak{b}_1 = 0$ , so  $\mathfrak{a}_1 \mathfrak{b}_1 = 0$ , and a fortiori

$$\mathfrak{a}_1 \mathfrak{a}_i = 0 \quad (i \neq 1);$$

$$\mathfrak{a}_1 = \mathfrak{a}_1 \mathfrak{o} = \mathfrak{a}_1 (\mathfrak{a}_1 + \mathfrak{b}_1) = (\mathfrak{a}_1^2, \mathfrak{a}_1 \mathfrak{b}_1) = \mathfrak{a}_1^2.$$

Conversely: from (1) and (2) it follows that the modules  $\mathfrak{a}_i$  are two-sided ideals. *Proof.*

$$\mathfrak{o} \mathfrak{a}_i = (\mathfrak{a}_1 + \cdots + \mathfrak{a}_n) \mathfrak{a}_i = \mathfrak{a}_i^2 = \mathfrak{a}_i, \quad \mathfrak{a}_i \mathfrak{o} = \mathfrak{a}_i.$$

*Right ideals in the ring  $\mathfrak{a}_i$  are right ideals in  $\mathfrak{o}$ . Proof.*

$$\mathfrak{r} \mathfrak{o} = \mathfrak{r} (\mathfrak{a}_i + \mathfrak{b}_i) = (\mathfrak{r} \mathfrak{a}_i, \mathfrak{r} \mathfrak{b}_i) = (\mathfrak{r}, 0) = \mathfrak{r}.$$

*If  $\mathfrak{r}$  is a right ideal in  $\mathfrak{o}$ , then  $\mathfrak{r}$  is a direct sum of right ideals  $\mathfrak{r} \mathfrak{a}_i \leq \mathfrak{a}_i$ . Proof.*  $\mathfrak{r} = \mathfrak{r} \mathfrak{o} = (\mathfrak{r} \mathfrak{a}_1, \dots, \mathfrak{r} \mathfrak{a}_n)$ , and  $\mathfrak{r} \mathfrak{a}_i \mathfrak{o} = \mathfrak{r} \mathfrak{a}_i$ , so the  $\mathfrak{r} \mathfrak{a}_i$  are right ideals contained in  $\mathfrak{r}$ . Since  $\mathfrak{r} \mathfrak{a}_i \leq \mathfrak{a}_i$ , the sum is direct:

$$\mathfrak{r} = \mathfrak{r} \mathfrak{a}_1 + \cdots + \mathfrak{r} \mathfrak{a}_n.$$

*In particular, if  $\mathfrak{r}$  is directly indecomposable, then  $\mathfrak{r}$  can have only one component; hence  $\mathfrak{r}$  must lie in one of the  $\mathfrak{a}_i$ .*

By means of these theorems, ideal theory in  $\mathfrak{o}$  is controlled once the ideal theory of the individual  $\mathfrak{a}_i$  is known. We shall therefore mostly restrict ourselves to two-sided indecomposable rings.

*Two right ideals contained in different  $\mathfrak{a}_i$  can never be operator-isomorphic. Proof.* An ideal lying in  $\mathfrak{a}_1$  is annihilated by all the other  $\mathfrak{a}_k$ , but not by  $\mathfrak{a}_1$ . By contrast, an ideal lying in  $\mathfrak{a}_2$  is annihilated by  $\mathfrak{a}_1$ .

Thus if it is possible to decompose the ring  $\mathfrak{o}$  into two-sided indecomposable ideals  $\mathfrak{a}_1 + \cdots + \mathfrak{a}_n$ , and each  $\mathfrak{a}_i$  again into one-sided indecomposable right ideals  $\mathfrak{r}'_i + \mathfrak{r}''_i + \cdots$ , then the operator-isomorphic  $\mathfrak{r}$  are to be sought among those belonging to the same  $\mathfrak{a}_i$ . But it may happen that there are still different classes of  $\mathfrak{r}$  – that is, classes not operator-isomorphic – in the same  $\mathfrak{a}_i$ , as the following example shows.

Let  $K$  be the field of rational numbers,

$$\mathfrak{o} = e_1 K + e_2 K + u K$$

a hypercomplex system, with the  $e_i$  and  $u$  commuting with the numbers of  $K$ , and with multiplication table

|       | $e_1$ | $e_2$ | $u$ |
|-------|-------|-------|-----|
| $e_1$ | $e_1$ | $0$   | $u$ |
| $e_2$ | $0$   | $e_2$ | $0$ |
| $u$   | $0$   | $u$   | $0$ |

The identity element is  $e = e_1 + e_2$ . The  $e_i$  satisfy the orthogonality relations. Thus the right decomposition is

$$\mathfrak{o} = e_1 \mathfrak{o} + e_2 \mathfrak{o} = (e_1, u) + (e_2),$$

and the left decomposition is

$$\mathfrak{o} = \mathfrak{o} e_1 + \mathfrak{o} e_2 = (e_1) + (e_2, u).$$

Since  $e_2 \mathfrak{o}$  and  $\mathfrak{o} e_1$  are visibly indecomposable,  $\mathfrak{o} e_2$  and  $e_1 \mathfrak{o}$  must be indecomposable as well.

A two-sided decomposition is impossible. For then one component would have to contain  $e_1 \mathfrak{o}$  and the other  $e_2 \mathfrak{o}$ ; the first would then contain  $u$ , but the second would contain  $u e_2 = u$  as well.

But  $e_1 \mathfrak{o}$  and  $e_2 \mathfrak{o}$  are not operator-isomorphic, since they have different rank with respect to  $K$ .

*Let  $\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_n$ , with the  $\mathfrak{a}_i$  two-sided indecomposable. Then they are uniquely determined.*

*Proof.* Suppose also  $\mathfrak{o} = \mathfrak{c}_1 + \cdots + \mathfrak{c}_n$ . Then

$$\mathfrak{a}_i = \mathfrak{a}_i \mathfrak{o} = (\mathfrak{a}_i \mathfrak{c}_1, \dots, \mathfrak{a}_i \mathfrak{c}_n).$$

This latter sum is direct because  $\mathfrak{a}_i \mathfrak{c}_k \leq \mathfrak{c}_k$  and  $\leq \mathfrak{a}_i$ ; since, however, the  $\mathfrak{a}_i$  are indecomposable, all  $\mathfrak{a}_i \mathfrak{c}_k$  must be zero except one,  $\mathfrak{a}_i \mathfrak{c}_{j_i}$ . Thus

$$\mathfrak{a}_i = \mathfrak{a}_i \mathfrak{c}_{j_i}.$$

Similarly,

$$\mathfrak{c}_{j_i} = \mathfrak{o} \mathfrak{c}_{j_i} = \mathfrak{a}_1 \mathfrak{c}_{j_i} + \cdots + \mathfrak{a}_n \mathfrak{c}_{j_i},$$

and since  $\mathfrak{a}_i \mathfrak{c}_{j_i} \neq 0$ , all the remaining terms are 0. Therefore

$$\mathfrak{c}_{j_i} = \mathfrak{a}_i \mathfrak{c}_{j_i} = \mathfrak{a}_i,$$

q.e.d.

## § 11. The center

The center  $\mathfrak{Z}$  of a ring  $\mathfrak{o}$  is the totality of all  $z$  commuting with all ring elements ( $za = az$  for all  $a$ ).  $\mathfrak{Z}$  is a commutative ring.

Every one-sided ideal  $\mathfrak{a}$  in  $\mathfrak{o}$  has a “contraction ideal”  $\mathfrak{a} \cap \mathfrak{Z}$  in  $\mathfrak{Z}$ . Since both  $\mathfrak{a}$  and  $\mathfrak{Z}$  are  $\mathfrak{Z}$ -modules, so is  $\mathfrak{a} \cap \mathfrak{Z}$ .

Every ideal  $\mathfrak{A}$  in  $\mathfrak{Z}$  has an extension ideal: the ideal  $\mathfrak{a} = (\mathfrak{A}, \mathfrak{o}\mathfrak{A})$  generated by  $\mathfrak{A}$  in  $\mathfrak{o}$ . This is two-sided, because

$$\mathfrak{a}\mathfrak{o} = (\mathfrak{o}\mathfrak{A}, \mathfrak{A})\mathfrak{o} = (\mathfrak{o}\mathfrak{A}\mathfrak{o}, \mathfrak{A}\mathfrak{o}) = (\mathfrak{o}^2\mathfrak{A}, \mathfrak{o}\mathfrak{A}) \subseteq \mathfrak{o}\mathfrak{A} \subseteq \mathfrak{a}.$$

If  $\mathfrak{o}$  is assumed to have an identity element, one may write  $\mathfrak{a} = \mathfrak{o}\mathfrak{A}$ . In this case the following theorem holds:

*Every two-sided decomposition of  $\mathfrak{o}$  corresponds one-to-one to a decomposition of the center: from*

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_n, \quad \mathfrak{A}_i = \mathfrak{a}_i \cap \mathfrak{Z} \tag{1}$$

*there follows*

$$\mathfrak{Z} = \mathfrak{A}_1 + \cdots + \mathfrak{A}_n, \quad \mathfrak{o}\mathfrak{A}_i = \mathfrak{a}_i, \tag{2}$$

*and conversely.*

*Proof.* Let  $z$  be a central element, and let

$$z = z_1 + \cdots + z_n \tag{3}$$

be its decomposition according to (1). Then the  $z_i$  are central elements; for

$$za = z_1 a + \cdots + z_n a = az = az_1 + \cdots + az_n.$$

By directness of the sum, and since  $\mathfrak{a}_i$  is two-sided, it follows that

$$z_i a = az_i.$$

Thus  $z_i$  lies in  $\mathfrak{a}_i \cap \mathfrak{Z} = \mathfrak{A}_i$ , so (3) gives the asserted sum decomposition. In particular,  $e = \sum e_i$ , hence the  $e_i$  are central elements. Finally

$$z = ze_1 + \cdots + ze_n,$$

and so

$$\mathfrak{A}_i = \mathfrak{Z}e_i, \quad \mathfrak{o}\mathfrak{A}_i = \mathfrak{o}\mathfrak{Z}e_i = \mathfrak{o}e_i = \mathfrak{a}_i.$$

Conversely, if  $\mathfrak{Z} = \mathfrak{A}_1 + \cdots + \mathfrak{A}_n$  is a decomposition of the center, then by §9

$$\mathfrak{A}_i = \mathfrak{Z}e_i, \quad e_i^2 = e_i, \quad e_i e_k = 0 \quad (i \neq k).$$

Hence, by the orthogonality relations §9,

$$\begin{aligned} \mathfrak{o} &= \mathfrak{o}e = \mathfrak{o}e_1 + \cdots + \mathfrak{o}e_n = \mathfrak{o}\mathfrak{Z}e_1 + \cdots + \mathfrak{o}\mathfrak{Z}e_n \\ &= \mathfrak{o}\mathfrak{A}_1 + \cdots + \mathfrak{o}\mathfrak{A}_n = \mathfrak{a}_1 + \cdots + \mathfrak{a}_n. \end{aligned}$$

If now  $\mathfrak{a}_i \cap \mathfrak{Z} = \mathfrak{A}'_i$ , then from the first part of the assertion one knows that

$$\mathfrak{Z} = \mathfrak{A}'_1 + \cdots + \mathfrak{A}'_n \quad \text{directly.}$$

But  $\mathfrak{A}'_i \supseteq \mathfrak{A}_i$ , hence  $\mathfrak{A}'_i = \mathfrak{A}_i$  by §4, 2.

*Corollary.*  $\mathfrak{a}_i$  and  $\mathfrak{A}_i$  are simultaneously two-sided directly decomposable or indecomposable.

## § 12. Nilpotent ideals

An ideal  $\mathfrak{c}$  is called *nilpotent* if  $\mathfrak{c}^\rho = 0$ .

*Example.* The ideal  $(u)$  in §10.

*If there is a nilpotent right ideal  $\mathfrak{r}$  in  $\mathfrak{o}$ , then there is also a nilpotent left ideal, indeed a two-sided ideal.*

*Proof.* Let  $\mathfrak{r}^\rho = 0$ . Then  $\mathfrak{o}\mathfrak{r}$ , or, if  $\mathfrak{o}$  contains no identity element and  $\mathfrak{o}\mathfrak{r}$  could be zero,  $(\mathfrak{o}\mathfrak{r}, \mathfrak{r})$ , is a left ideal, and

$$(\mathfrak{o}\mathfrak{r})^\rho = \mathfrak{o}\mathfrak{r}\mathfrak{o}\mathfrak{r}\cdots\mathfrak{o}\mathfrak{r} \subseteq \mathfrak{o}\mathfrak{r}\mathfrak{r}\cdots\mathfrak{r} = \mathfrak{o}\mathfrak{r}^\rho = 0.$$

*The sum of two nilpotent right ideals is again a nilpotent right ideal.*

*Proof.* Let  $\mathfrak{c}^\rho = 0$ ,  $\mathfrak{d}^\sigma = 0$ . In

$$(\mathfrak{c}, \mathfrak{d})^{\rho+\sigma-1} = (\dots, \mathfrak{d}\mathfrak{c}\cdots\mathfrak{d}\cdots\mathfrak{c}, \dots)$$

every term contains either at least  $\rho$  factors  $\mathfrak{c}$  or at least  $\sigma$  factors  $\mathfrak{d}$ . In the first case one has

$$\mathfrak{d}\mathfrak{c}\cdots\mathfrak{d}\cdots\mathfrak{c}\cdots \subseteq \mathfrak{d}\mathfrak{c}\mathfrak{c}\cdots\mathfrak{c} = \mathfrak{d}\mathfrak{c}^\rho = 0;$$

in the latter case the same argument applies to the factors  $\mathfrak{d}$ . Thus

$$(\mathfrak{c}, \mathfrak{d})^{\rho+\sigma-1} = 0.$$

If now the maximal condition for right ideals in  $\mathfrak{o}$  is satisfied, then there is a maximal nilpotent right ideal. It contains all other nilpotent right ideals; otherwise one could form its sum with such an ideal. Call it  $\mathfrak{c}$ . Since  $\mathfrak{o}\mathfrak{c}$  is also a nilpotent right ideal,  $\mathfrak{o}\mathfrak{c} \subseteq \mathfrak{c}$ , so  $\mathfrak{c}$  is a left ideal. Every nilpotent left ideal  $\mathfrak{d}$  is likewise contained in  $\mathfrak{c}$ , since  $(\mathfrak{d}, \mathfrak{d}\mathfrak{o})$  is a nilpotent right ideal. Thus  $\mathfrak{d} \subseteq \mathfrak{c}$ . *There is therefore a maximal two-sided nilpotent ideal  $\mathfrak{c}$ , or “radical,” containing all the others, right- and left-sided.*

In the example of §10,  $(u)$  is the maximal nilpotent ideal.

If the radical is the zero ideal, one speaks of a “ring without radical” (“semisimple ring,” “Dedekind system”). The residue-class ring modulo the radical is always a ring without radical.

*If  $\mathfrak{Z}$  is the center of  $\mathfrak{o}$ , and  $\mathfrak{c}$  is the radical of  $\mathfrak{o}$ , then  $\mathfrak{C} = \mathfrak{c} \cap \mathfrak{Z}$  is the radical of  $\mathfrak{Z}$ .*

*Proof.* It is clear that  $\mathfrak{C}$  is nilpotent. If there were a larger nilpotent ideal  $\bar{\mathfrak{C}}$  in  $\mathfrak{Z}$ , then it could be extended to a nilpotent ideal  $\bar{\mathfrak{c}}$  not wholly contained in  $\mathfrak{c}$ , since  $((\bar{\mathfrak{C}}\mathfrak{o})^\rho = \bar{\mathfrak{C}}^\rho \mathfrak{o}^\rho = 0)$ .

*Consequence.* If  $\mathfrak{o}$  is a ring without radical, so is  $\mathfrak{Z}$ . The converse is not true, however, as the example in §10 shows. The center of  $\mathfrak{o}/\mathfrak{c}$  contains a ring isomorphic to  $\mathfrak{Z}/\mathfrak{C}$ , but this can be a proper subring (example in §10).

## § 13. Completely reducible rings

By §5 a ring is called right completely reducible if it is a direct sum of finitely many simple right ideals.

*A right completely reducible ring with identity element has no nilpotent ideal  $\neq (0)$ , hence no radical.*

*Proof.* Every right ideal  $\mathfrak{r}$  is a direct summand (§5):

$$\mathfrak{o} = \mathfrak{t} + \mathfrak{r} = e_1\mathfrak{o} + e_2\mathfrak{o}.$$

Since  $e_2^2 = e_2$ , also  $e_2^\rho = e_2$ . If  $\mathfrak{r}$  were nilpotent, then  $e_2^\rho = 0$ , hence  $e_2 = 0$ , and so  $\mathfrak{r} = 0$ , q.e.d.

We now show the two converses. First: *A ring without radical satisfying the minimal condition for right ideals is completely reducible with respect to right ideals.*

*Proof.* Let  $\mathfrak{t}$  be a minimal right ideal  $\neq 0$ . We shall show that  $\mathfrak{t}$  contains an idempotent element.

Since  $\mathfrak{t}^2 \subseteq \mathfrak{t}$  and  $\mathfrak{t}^2 \neq 0$ , one has  $\mathfrak{t}^2 = \mathfrak{t}$ , for  $\mathfrak{t}^2$  is a right ideal. Hence there is an  $a$  in  $\mathfrak{t}$  such that  $a\mathfrak{t} \neq 0$ . Then necessarily  $a\mathfrak{t} = \mathfrak{t}$ , because  $a\mathfrak{t}$  is a right ideal.

The totality of all  $b$  in  $\mathfrak{t}$  which are annihilated by  $a$  ( $ab = 0$ ) is a right ideal, and is not equal to  $\mathfrak{t}$ , hence is 0. Thus  $ab = 0$  implies  $b = 0$ .

Because  $\mathfrak{t} = a\mathfrak{t}$ ,  $a$  must have a representation  $a = ac$ , with  $c \neq 0$ . It follows that

$$ac = ac^2, \quad a(c - c^2) = 0, \quad c - c^2 = 0, \quad c^2 = c.$$

We next show that  $\mathfrak{t}$  is a direct summand. The set  $c\mathfrak{o}$  is a non-zero right ideal contained in  $\mathfrak{t}$ , since  $c^2 = c$  lies in  $c\mathfrak{o}$ ; hence it is  $\mathfrak{t}$ . Every element  $r$  of  $\mathfrak{o}$  has a representation

$$r = cr + (r - cr) \quad (\text{one-sided Peirce decomposition}).$$

The elements  $cr$  form the ideal  $\mathfrak{t}$ ; the elements  $r - cr$  form another right ideal  $\mathfrak{r}$ , annihilated by  $c$ :  $c\mathfrak{r} = 0$ . Hence

$$\mathfrak{o} = (\mathfrak{t}, \mathfrak{r}).$$

Since the elements of  $\mathfrak{t}$  are not annihilated by  $c$ , the sum is direct:

$$\mathfrak{o} = \mathfrak{t} + \mathfrak{r}. \quad (1)$$

Representing  $\mathfrak{o}$ , by the minimal condition, as a direct sum of directly indecomposable ideals,

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n,$$

the  $\mathfrak{r}_i$  must be simple, i.e. minimal. For if, say,  $\mathfrak{r}_1$  were not simple and  $\mathfrak{t}$  were a minimal ideal in  $\mathfrak{r}_1$ , then (1) would give

$$\mathfrak{r}_1 = \mathfrak{t} + \mathfrak{r} \cap \mathfrak{r}_1 \quad \S 4, 3,$$

so  $\mathfrak{r}_1$  would be decomposable, a contradiction. Thus  $\mathfrak{o}$  is completely reducible.

Second: *A ring without radical satisfying the minimal condition has an identity element.*

To show first the existence of a left identity, it is enough to prove the following: *if a right ideal  $\mathfrak{a} = c_1\mathfrak{o} \neq \mathfrak{o}$  has a left identity  $c_1$ , then there is a right ideal  $c\mathfrak{o}$  of greater length which likewise has a left identity  $c$ .* For, since complete reducibility, and therefore boundedness of all lengths, has already been shown, one reaches in finitely many steps a left identity for  $\mathfrak{o}$  itself.

Let therefore  $\mathfrak{a} = c_1\mathfrak{o}$  and  $c_1^2 = c_1$ . By the formula

$$r = c_1r + (r - c_1r)$$

we have, as above, a Peirce decomposition  $\mathfrak{o} = c_1\mathfrak{o} + \mathfrak{r}$ . Let  $c_2$  be an idempotent element of  $\mathfrak{r}$ , as above; thus  $c_2^2 = c_2$ , and  $c_1c_2 = 0$  since  $c_1$  annihilates all elements of  $\mathfrak{r}$ . Put  $e_1 = c_1$ ,  $e_2 = c_2 - c_2c_1$ . Then

$$e_1^2 = e_1, \quad e_1e_2 = 0, \quad e_2e_1 = 0, \quad e_2c_2 = e_2, \quad e_2 \neq 0, \quad e_2^2 = e_2.$$

By the remark made in the note above,  $c = e_1 + e_2$  is a left identity for the ring

$$c\mathfrak{o} = e_1\mathfrak{o} + e_2\mathfrak{o},$$

which, since  $e_2^2 = e_2 \neq 0$ , has greater length than  $\mathfrak{a}$  as a right ideal.

To show that the constructed left identity  $e$  is also a right identity, we make the Peirce decomposition into left ideals:

$$\mathfrak{o} = \mathfrak{o}e + \mathfrak{l}.$$

We have  $\mathfrak{l}e = 0$ ,  $\mathfrak{l} = e\mathfrak{l}$ , hence  $\mathfrak{l}^2 = \mathfrak{l}e\mathfrak{l} = 0$ . Since by hypothesis no nilpotent left ideal can exist,  $\mathfrak{l} = 0$ . Therefore  $e$  is also a right identity, and hence an identity altogether.

*Theorem. From right-sided complete reducibility and the existence of the identity follows two-sided complete reducibility:*

$$\mathfrak{o} = \mathfrak{d}_1 + \cdots + \mathfrak{d}_s, \quad \mathfrak{d}_i \text{ two-sided simple.}$$

As in the preceding proof, it is enough to show that every minimal two-sided ideal  $\mathfrak{a}$  is a direct summand. As a right ideal,  $\mathfrak{a}$  is a direct summand:

$$\mathfrak{o} = \mathfrak{a} + \mathfrak{r} = e_1\mathfrak{o} + e_2\mathfrak{o}.$$

The corresponding left decomposition is

$$\mathfrak{o} = \mathfrak{c} + \mathfrak{l} = \mathfrak{o}e_1 + \mathfrak{o}e_2.$$

It follows that

$$\begin{aligned} \mathfrak{r}\mathfrak{c} &\subseteq \mathfrak{r}\mathfrak{a} \leq \mathfrak{r} \cap \mathfrak{a} = 0, & \mathfrak{l}\mathfrak{a} &= \mathfrak{o}e_2e_1\mathfrak{o} = 0, \\ (\mathfrak{a}\mathfrak{l})^2 &= \mathfrak{a}\mathfrak{l}\mathfrak{a}\mathfrak{l} = 0, & \text{hence } \mathfrak{a}\mathfrak{l} &= 0; \\ \mathfrak{r} &= \mathfrak{r}\mathfrak{o} = (\mathfrak{r}\mathfrak{c}, \mathfrak{r}\mathfrak{l}) = \mathfrak{r}\mathfrak{l}, & \mathfrak{l} &= \mathfrak{o}\mathfrak{l} = (\mathfrak{a}\mathfrak{l}, \mathfrak{r}\mathfrak{l}) = \mathfrak{r}\mathfrak{l}. \end{aligned}$$

Thus  $\mathfrak{r} = \mathfrak{l}$ , so  $\mathfrak{r}$  is two-sided; hence  $\mathfrak{a}$  is a two-sided direct summand. From here on the proof is the same as for one-sided complete reducibility.

The  $\mathfrak{d}_i$  are also two-sided simple as rings, since all ideals in  $\mathfrak{d}_i$  are ideals in  $\mathfrak{o}$ . We now investigate their structure.



## § 14. Two-sided simple completely reducible rings with identity element

Let  $\mathfrak{o}$  be such a ring:

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n = e_1 \mathfrak{o} + \cdots + e_n \mathfrak{o}, \quad \mathfrak{r}_i \text{ simple.}$$

Then

$$\mathfrak{o} = \mathfrak{l}_1 + \cdots + \mathfrak{l}_n = \mathfrak{o}e_1 + \cdots + \mathfrak{o}e_n, \quad \mathfrak{l}_i \text{ directly indecomposable §9.}$$

Moreover  $\mathfrak{l}_i \mathfrak{r}_i$  is a non-zero two-sided ideal (since  $e_i^2$  lies in  $\mathfrak{l}_i \mathfrak{r}_i$ ), hence is  $\mathfrak{o}$ .

Put  $\mathfrak{A}_{ik} = \mathfrak{r}_i \mathfrak{l}_k$ . Then

$$\mathfrak{o} = (\mathfrak{r}_1, \dots, \mathfrak{r}_n) \cdot (\mathfrak{l}_1, \dots, \mathfrak{l}_n) = (\dots, \mathfrak{A}_{ij}, \dots)$$

directly. For if  $0 = \sum_{i,j} a_{ij}$ , then, since  $a_{ij}$  lies in  $\mathfrak{r}_i$  and  $\sum \mathfrak{r}_i$  is direct,

$$0 = \sum_j a_{ij},$$

and then, because  $a_{ij}$  lies in  $\mathfrak{l}_j$ ,

$$0 = a_{ij}.$$

Further,  $\mathfrak{A}_{ij} \mathfrak{r}_j = \mathfrak{r}_i \mathfrak{l}_j \mathfrak{r}_j = \mathfrak{r}_i \mathfrak{o} = \mathfrak{r}_i$ , so  $\mathfrak{A}_{ij} \neq 0$ . Let  $a_{ij} \neq 0$  lie in  $\mathfrak{A}_{ij}$ . Then

$$a_{ij} \mathfrak{r}_j \subseteq \mathfrak{r}_i, \quad a_{ij} \mathfrak{r}_j \neq 0 \text{ because } a_{ij} e_j = a_{ij},$$

and therefore

$$a_{ij} \mathfrak{r}_j = \mathfrak{r}_i.$$

If for every  $r_j$  in  $\mathfrak{r}_j$  one puts

$$a_{ij} r_j = r_i,$$

then the map  $r_j \mapsto r_i$  is an operator homomorphism from  $\mathfrak{r}_j$  to  $\mathfrak{r}_i$ . The elements mapped to zero form an ideal contained in  $\mathfrak{r}_j$ , hence the zero ideal; therefore the map is an *isomorphism*. Thus:

*Any two simple right ideals  $\mathfrak{r}_i$  and  $\mathfrak{r}_j$  occurring in a representation of  $\mathfrak{o}$  are operator-isomorphic; the isomorphisms are mediated by the elements of  $\mathfrak{A}_{ij}$ .* Since any two different simple right ideals  $\mathfrak{r}, \mathfrak{r}'$  occur together in at least one representation of  $\mathfrak{o}$ , every two such ideals are isomorphic.

*All homomorphisms from  $\mathfrak{r}_j$  to  $\mathfrak{r}_i$  are mediated by elements of  $\mathfrak{A}_{ij}$ .*

*Proof.* If  $e_j \mapsto a_{ij}$ , then  $e_j^2 \mapsto a_{ij} e_j$ , so  $a_{ij} = a_{ij} e_j$  lies in  $\mathfrak{r}_i \mathfrak{l}_j = \mathfrak{A}_{ij}$ . Further,

$$\mathfrak{r}_j = e_j \mathfrak{r}_j \mapsto a_{ij} \mathfrak{r}_j = \mathfrak{r}_i.$$

*In particular, the homomorphisms of  $\mathfrak{r}_i$  into itself are mediated by  $\mathfrak{A}_{ii}$ .*

The product of two elements of  $\mathfrak{A}_{ii}$  corresponds to the product of the homomorphisms, and the sum to the sum (definition: see §1, 6). Distinct  $a_{ii}$  give distinct automorphisms, for  $e_i$  goes to  $a_{ii} e_i = a_{ii}$ . Thus the ring  $\mathfrak{A}_{ii}$  is isomorphic to the automorphism ring of  $\mathfrak{r}_i$ .

But the automorphism ring of a simple ideal is a field (§2), hence  $\mathfrak{A}_{ii}$  is a field. Since all  $\mathfrak{r}_i$  are operator-isomorphic, their automorphism rings are ring-isomorphic:  $\mathfrak{A}_{ii}$  is uniquely determined by  $\mathfrak{o}$  up to ring isomorphism. The possible different  $\mathfrak{A}_{ii}$  are only different concrete realizations of the abstractly defined automorphism ring of the simple right ideals.

*Main theorem. Every completely reducible two-sided simple ring  $\mathfrak{o}$  with identity element is isomorphic to the ring of matrices of degree  $n$  over a field  $K$ . (The field  $K$  is isomorphic to the automorphism field of the right ideals of  $\mathfrak{o}$ .)*

*Proof. Construction of the matrix units  $c_{ik}$ .* Let  $\Gamma_{11}$  be the identical automorphism of  $\mathfrak{r}_1$ , let  $\Gamma_{i1}$  be an arbitrary isomorphism  $\Gamma_{i1} \mathfrak{r}_1 = \mathfrak{r}_i$ , and finally set

$$\Gamma_{ik} = \Gamma_{i1} \Gamma_{k1}^{-1}.$$

Then in general

$$\Gamma_{ik} \Gamma_{kl} = \Gamma_{il}.$$

If  $\Gamma_{ik}$  is mediated by the element  $c_{ik}$  of  $\mathfrak{A}_{ik}$ , then further

$$c_{ik} = e_i c_{ik} = c_{ik} e_k, \quad c_{ik} c_{kl} e_l = c_{il} e_l,$$

hence

$$c_{ik}c_{kl} = c_{il}, \quad c_{ij}c_{kl} = c_{ij}e_j e_k c_{kl} = 0 \quad (j \neq k).$$

The  $c_{ik}$  are therefore matrix units (§7).

*Construction of  $K$ .* The assignment

$$a_{ij} = c_{i1}a_{11}c_{1i}$$

assigns to every  $a_{11}$  in  $\mathfrak{A}_{11}$  a “conjugate element”  $a_{ii}$  of  $\mathfrak{A}_{ii}$ .

This assignment is a ring isomorphism, since the isomorphisms  $\Delta, \Gamma$  corresponding to  $a$  and  $c$  are connected by

$$\Delta_{ii} = \Gamma_{i1}\Delta_{11}\Gamma_{i1}^{-1},$$

a relation well known to represent a ring isomorphism. Now form the elements

$$\alpha = a_{11} + \cdots + a_{nn} = a_{11} + c_{21}a_{11}c_{12} + \cdots + c_{n1}a_{11}c_{1n}.$$

Each  $a_{11}$  determines exactly one  $\alpha$  (uniquely because the sum is direct). The totality of the  $\alpha$  is a field  $K \sim \mathfrak{A}_{11}$ , since from

$$\alpha = a_{11} + \cdots + a_{nn}, \quad \beta = b_{11} + \cdots + b_{nn},$$

one obtains

$$\alpha + \beta = (a_{11} + b_{11}) + \cdots + (a_{nn} + b_{nn}), \quad \alpha\beta = a_{11}b_{11} + \cdots + a_{nn}b_{nn};$$

so the assignment  $a_{11} \mapsto \alpha$  is an isomorphism.

Further,

$$c_{ik}\alpha = c_{ik}a_{kk} = c_{ik}c_{k1}a_{11}c_{1k} = c_{i1}a_{11}c_{1k},$$

$$\alpha c_{ik} = a_{ii}c_{ik} = c_{i1}a_{11}c_{1i}c_{ik} = c_{i1}a_{11}c_{1k},$$

so every  $\alpha$  commutes with all  $c_{ik}$ .

Finally, the same formulas give

$$c_{ik}K = c_{ik}\mathfrak{A}_{kk} = c_{ik}\mathfrak{r}_k\mathfrak{l}_k = \mathfrak{r}_i\mathfrak{l}_k = \mathfrak{A}_{ik},$$

and therefore

$$\mathfrak{o} = \sum c_{ik}K.$$

*The assignment.* If every element of  $\mathfrak{o}$  is written in the form  $\sum c_{ik}\alpha_{ik}$ , and the matrix  $(\alpha_{ik})$  is assigned to it, the assignment is an isomorphism, as follows immediately from the definition of matrix multiplication. This proves the main theorem.

*Converse.* The ring of matrices of degree  $n$  over a field  $A$  is two-sided simple and completely reducible;  $A$  is isomorphic to the automorphism ring of the right ideals, and  $Ae$  is the  $K$  constructed in the preceding theorem.

*Proof.* Let the matrix units (§7) be  $c_{ik}$ . Put

$$\mathfrak{r}_i = \sum_k c_{ik}A.$$

Then plainly  $\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n$ . The  $\mathfrak{r}_i$  are simple right ideals, for every non-zero element  $\sum c_{ik}\alpha_k$  generates the full  $\mathfrak{r}_i$ . Indeed, if  $\alpha_j \neq 0$ , then

$$\left(\sum c_{ik}\alpha_k\right) \cdot (c_{jl}\alpha_j^{-1}) = c_{il},$$

and  $\sum c_{il}\beta_l$  runs through all elements of  $\mathfrak{r}_i$ .

The  $\mathfrak{r}_i$  are operator-isomorphic:  $\mathfrak{r}_i = c_{ik}\mathfrak{r}_k$ .

It follows that  $\mathfrak{o}$  is two-sided indecomposable; hence, by the theorem of §13 and the already proved complete reducibility, and since an identity element exists, it is two-sided simple. This is also seen from the fact that any  $\sum \sum c_{ik}\alpha_{ik} \neq 0$  generates the entire two-sided ideal  $\mathfrak{o}$ .

Further,

$$\begin{aligned} \mathfrak{r}_i &= c_{ii}\mathfrak{o}, & \mathfrak{l}_i &= \mathfrak{o}c_{ii}, \\ e &= \sum c_{ii}, & c_{ii} &= e_i, \end{aligned}$$

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<sup>0</sup>Recently B. L. van der Waerden found a simpler proof, operating directly with the abstract automorphism field, which is to appear in his book on algebra (Grundlehren d. Math. Wiss., Berlin: Julius Springer). [11 June 1929.]

$$\mathfrak{A}_{ii} = \mathfrak{r}_i \cap \mathfrak{l}_i = c_{ii}A \sim A.$$

Finally, putting  $a_{11} = c_{11}\lambda$ , one obtains

$$\alpha = a_{11} + c_{21}a_{11}c_{12} + \cdots = c_{11}\lambda + c_{22}\lambda + \cdots = e\lambda,$$

which proves everything asserted.

*The center of  $\mathfrak{o}$  is the center of  $K$ , hence a field.*

*Proof.*  $\mathfrak{Z}(K)$  consists of all  $\zeta$  that commute with all  $x$ . But these also commute with all  $c_{ik}$ , and hence with all sums  $\sum c_{ik}a_{ik}$ . Thus

$$\mathfrak{Z}(K) \subseteq \mathfrak{Z}(\mathfrak{o}).$$

Conversely, let  $z$  lie in  $\mathfrak{Z}(\mathfrak{o})$ :

$$z = \sum_{\mu} \sum_{\nu} c_{\mu\nu} \gamma_{\mu\nu}.$$

Then

$$zc_{ik} = c_{ik}z, \quad \sum_{\mu} c_{\mu k} \gamma_{\mu i} = \sum_{\nu} c_{i\nu} \gamma_{k\nu},$$

and so

$$\begin{aligned} \gamma_{\mu i} &= 0 \quad (\mu \neq i), & c_{ik} \gamma_{ii} &= c_{ik} \gamma_{kk}, & \gamma_{ii} &= \gamma_{kk} = \gamma; \\ z &= \sum c_{ii} \gamma = e\gamma = \gamma. \end{aligned}$$

Thus  $z$  lies in  $K$  and commutes with all  $x$ , hence  $z$  lies in  $\mathfrak{Z}(K)$ .

Since a matrix ring is of course also left completely reducible, and since every right completely reducible ring with identity element is a direct sum of matrix rings, it follows that:

*Every right completely reducible ring with identity element is also left completely reducible, and conversely.*

*And: The center of a completely reducible ring with identity element is a direct sum of commutative fields corresponding to the two-sided simple matrix rings.*

There remains the following theorem:

*Any two different decompositions  $\mathfrak{o} = \sum c_{ik}K$ ,  $\mathfrak{o} = \sum c'_{ik}K'$  are transformed into one another by inner automorphisms  $\alpha' \mapsto x^{-1}\alpha'x$ ,  $c'_{ik} \mapsto x^{-1}c_{ik}x$ .*

*Proof.* Let

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n = \mathfrak{r}'_1 + \cdots + \mathfrak{r}'_n,$$

and let  $a, b$  be elements mediating two reciprocal isomorphisms of the ideals  $\mathfrak{r}_1, \mathfrak{r}'_1$ :

$$\mathfrak{r}_1 = a\mathfrak{r}'_1, \quad \mathfrak{r}'_1 = b\mathfrak{r}_1, \quad ab = c_{11}, \quad ba = c'_{11}.$$

Put

$$x = \sum_i c_{i1}ac'_{1i}, \quad y = \sum_i c'_{i1}bc_{1i}.$$

Then

$$xy = e, \quad y = x^{-1}, \quad x^{-1}c_{ik}x = c'_{ik}, \quad x^{-1}\alpha x = \alpha'.$$